Scaling Book Exercises: Section 3

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Chapter: Sharded Matrices and How to Multiply Them
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Notation. W denotes bidirectional ICI bandwidth unless stated otherwise, C denotes per-accelerator FLOPs/s, and X, Y, Z denote mesh-axis sizes as well as their axis names when the meaning is clear. Arrays are in bfloat16 unless specified, so each element occupies two bytes.

Exercise 1

The mesh is $\{X : 4, Y : 8, Z : 2\}$ and $A[I_X, J, K, \ldots]$ is sharded only over X. Each device stores one quarter of A, while the data is replicated over Y and Z. Across all 64 devices,

$$64\frac{|A|}{4} = 16|A|.$$

Thus the ratio of total bytes to one logical copy is

$$\boxed{16}.$$

Exercise 2

For $B = 1024$, $D = 4096$, mesh $\{X : 4, Y : 4, Z : 4\}$, and TPU v4p bandwidth $W = 9.0 \times 10^{10}$ bytes/s:

(a) Gathering $[B_X, D_Y]$ over X moves the Y-sharded fraction of the array:

$$T_{\text{AllGather}_X} = \frac{2BD}{YW} = \frac{2(1024)(4096)}{4(9.0 \times 10^{10})} \approx \boxed{23\ \mu\text{s}}.$$

(b) Gathering over both X and Y moves the full array while using two mesh axes:

$$T_{\text{AllGather}_{XY}} = \frac{2BD}{2W} \approx \boxed{46\ \mu\text{s}}.$$

(c) An AllReduce is a ReduceScatter followed by an AllGather. The local output shard has $2BD/(XY)$ bytes, so

$$T_{\text{AllReduce}_Z} = \frac{4BD}{XYW} \approx \boxed{11.6\ \mu\text{s}}.$$

Exercise 3

$[B_X]$ contains only $2B = 256$ bytes globally and 64 bytes per device. Its bandwidth time is negligible, but a bidirectional gather around an axis of size four needs two hops. At roughly 1 μs per hop,

$$\boxed{T \approx 2\ \mu\text{s}}.$$

Exercise 4

For $X[B, D]\, Y[D_X, F] \to Z[B, F]$, compare two strategies.

Strategy 1: gather weights, then multiply.

$$T_1 = \max\left(\frac{2BDF}{C}, \frac{2DF}{W}\right).$$

It performs $2BDF$ FLOPs per device and communicates $2DF$ bytes.

Strategy 2: local multiply, then AllReduce.

$$T_2 = \max\left(\frac{2BDF}{XC}, \frac{4BF}{W}\right).$$

It performs $2BDF/X$ FLOPs per device and AllReduces a $2BF$-byte output.

Strategy 2 is compute-bound only if

$$X < \frac{DW}{2C}.$$

With TPU v5p values $C/W \approx 2550$ and $D \approx 8000$, this requires $X < 1.6$, so Strategy 2 is normally communication-bound. Strategy 1 becomes compute-bound at $B > C/W \approx 2550$. If $B < 2550$, Strategy 2 wins when $D > 2B$; for large B, it wins when $D > 2C/W \approx 5100$. Hence the local-matmul/AllReduce strategy is usually better for large hidden dimensions.

Exercise 5

On a $4 \times 4 \times 4$ mesh, let $P = XYZ = 64$. To minimize latency while returning a replicated result, shard one non-contracting output dimension, perform the local matmul, then AllGather the output. Two symmetric choices are

$$A[I_{XYZ}, J]B[J, K] \to C[I_{XYZ}, K] \xrightarrow{\text{AllGather}_{XYZ}} C[I, K]$$

or

$$A[I, J]B[J, K_{XYZ}] \to C[I, K_{XYZ}] \xrightarrow{\text{AllGather}_{XYZ}} C[I, K].$$

For either choice,

$$T = \max\left(\frac{2IJK}{64C}, \frac{2IK}{3W}\right).$$

When compute-bound, the calculation is 64 times faster than full replication. In the communication-bound case, it still beats full replication when

$$J > \frac{C}{3W}.$$

Using the scan's $C = 2.75 \times 10^{14}$ and $W = 9.0 \times 10^{10}$ gives a crossover near $J \approx 1019$.

Exercise 6

Consider a TPU v5e 4×4 mesh.

(a) For

$$A[I_X, J_Y]B[J_Y, K] \to C[I_X, K],$$

both contracting axes are sharded identically. Local matmuls produce partial sums unreduced over Y, followed by AllReduce_Y:

$$T_{\text{math}} = \frac{2IJK}{XYC}, \qquad T_{\text{comm}} = \frac{4IK}{XW}.$$

(b) For

$$A[I_X, J]B[J_X, K_Y] \to C[I_X, K_Y],$$

the shared use of mesh axis X conflicts. First AllGather_X the J_X dimension of B, then perform the local matmul:

$$T_{\text{math}} = \frac{2IJK}{XYC}, \qquad T_{\text{comm}} = \frac{2JK}{YW}.$$

(c) For

$$A[I_X, J]B[J, K_Y] \to C[I_X, K_Y],$$

neither contracting dimension is sharded and the two non-contracting dimensions use different mesh axes. No communication is required:

$$T_{\text{math}} = \frac{2IJK}{XYC}, \qquad T_{\text{comm}} = 0.$$

Exercise 7

W_{in} and W_{out} each occupy

$$2DF = 2(8192)(32768) \approx 537 \text{ MB},$$

so they must be sharded across the four-device 2×2 slice. A tensor- parallel layout is

$$\text{In}[B, D_{XY}]\, W_{\text{in}}[D, F_{XY}] \to H[B, F_{XY}],$$

$$H[B, F_{XY}]\, W_{\text{out}}[F_{XY}, D] \to \text{Out}[B, D]\{U_{XY}\} \xrightarrow{\text{ReduceScatter}_{XY,D}} \text{Out}[B, D_{XY}].$$

This uses an initial AllGather_{XY} of the input and a final $\text{ReduceScatter}_{XY}$ of the output. The two matmuls perform

$$\frac{4BDF}{XY}$$

FLOPs per device, while total communication time is

$$T_{\text{comm}} = \frac{2BD}{W}.$$

With $B = 128$, $D = 8192$, $F = 32768$, $XY = 4$, $C = 1.97 \times 10^{14}$ FLOPs/s, and $W = 9.0 \times 10^{10}$ bytes/s,

$$T_{\text{math}} \approx 174\ \mu\text{s}, \qquad T_{\text{comm}} \approx 23.3\ \mu\text{s}.$$

With overlap, the estimate is therefore $\boxed{174\ \mu\text{s}}$; without overlap, approximately 197 μs.

Exercise 8

The scan records the JAX collective mappings:

$$\begin{aligned}\text{AllGather} &: \texttt{jax.lax.all\_gather},\\ \text{AllReduce} &: \texttt{jax.lax.psum},\\ \text{ReduceScatter} &: \texttt{jax.lax.psum\_scatter},\\ \text{AllToAll} &: \texttt{jax.lax.all\_to\_all}.\end{aligned}$$

Each benchmark should be mapped over a named axis. Synchronize every result with

```
jax.block_until_ready(result)
```

so dispatch latency is not mistaken for collective runtime. The handwritten work does not include measured timing results, so this exercise remains an implementation benchmark rather than a numerically verified result.

Exercise 9

Let device r own contracting-index set S_r. It computes the partial result

$$C_r[n,k] = \sum_{m \in S_r} A[n,m]B[m,k].$$

An AllReduce forms the replicated product:

$$C[n,k] = \sum_r C_r[n,k] = \sum_{m=0}^{M-1} A[n,m]B[m,k].$$

If a sharded output is acceptable, replace the AllReduce with a ReduceScatter along either N or K. Gathering $A[N,M]$ communicates an amount proportional to NM; reduce-scattering $C[N,K]$ communicates an amount proportional to NK. Therefore,

$$\frac{\text{gather-first cost}}{\text{matmul-first cost}} = \boxed{\frac{M}{K}}.$$

Gather first when $M < K$, reduce-scatter after the matmul when $M > K$, and either when $M = K$.

Exercise 10

Let D devices form a ring and let the logical matrix contain N^2 scalars.

(a) On a unidirectional ring, AllGather or ReduceScatter sends a strip of size N^2/D for $D-1$ steps. Per link,

$$V_{\text{AG,uni}} = N^2 \frac{D-1}{D} \longrightarrow N^2.$$

(b) For AllToAll, destination blocks shrink along the route:

$$V_{\text{ATA,uni}} = \frac{N^2}{D^2} \sum_{j=1}^{D-1} j = \frac{N^2(D-1)}{2D} \longrightarrow \frac{N^2}{2}.$$

(c) Thus unidirectional AllToAll costs half as much as AllGather.

(d) Bidirectionality halves AllGather time because each strip travels in both directions.

(e) Bidirectionality reduces AllToAll time by a factor of four: blocks choose the shorter direction and travel at most $\lfloor D/2 \rfloor$ hops.

(f) Combining the ratios,

$$\frac{T_{\text{AG,bi}}}{T_{\text{ATA,bi}}} = \frac{(1/2)T_{\text{AG,uni}}}{(1/4)T_{\text{ATA,uni}}} = \boxed{4}.$$